Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 01)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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8	
9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Table 1 shows the number of adult great crested newts caught in bottle traps in a pond over five years, and the number of ponds in the area in which the species was detected using environmental DNA (eDNA).

Table 1

Year	Adults caught in traps	Ponds with newts detected by eDNA (out of 20)
2019	14	6
2020	22	9
2021	18	11
2022	31	13
2023	27	15

0 1 . 1 Calculate the percentage increase in the number of ponds where newts were detected between 2019 and 2023.

[2 marks]

Answer ..... %

0 1 . 2 Explain how eDNA detects the presence of newts without catching them.

[2 marks]

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0 1 . 3 Suggest why trap catches are a less reliable measure of population change than the eDNA survey.

[2 marks]

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Do not write outside the box

0 1 . 4 Evaluate the contribution of new technologies to ecological monitoring and wildlife conservation.

[9 marks]

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Extra space

15

0 2 . 1 Explain how GPS mapping, drone surveys and computer-based yield monitoring can reduce the environmental impacts of agriculture.

[3 marks]

0 2 . 2 Explain why productive agricultural technologies are less available in LEDCs.

[2 marks]

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0 2 . 3 Explain how buying fairtrade food affects producers.

[1 mark]

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0 2 . 4 Explain how nitrate from farmland can threaten human health.

[2 marks]

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0 2 . 5 Give **two** agricultural sources of methane.

[2 marks]

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2 .....

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0 3 Table 1 shows some medicines that were discovered in wild species.

Table 1

Medicine	Source species	Use
Taxol	Yew tree	Cancers of the breast, ovary, lung, bladder and prostate
Quinine	Cinchona tree	Malaria
AZT	Caribbean marine sponge	HIV / AIDS
Diosgenin (steroids)	Mexican yam	Contraceptive pill, cortisone

0 3 . 1 Explain why many plants produce chemicals that turn out to be useful medicines.

[2 marks]

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0 3 . 2 Explain why the medicines in Table 1 provide an argument for conserving whole habitats rather than individual species.

[2 marks]

0 3 . 3 Describe how **one** named species is being studied to help solve a human health problem other than by providing a medicine.

[2 marks]

0 3 . 4 Discuss the extent to which the resources and ecosystem services provided by other species justify the conservation of biodiversity.

[9 marks]

Do not write outside the box

Extra space

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15

0 4 . 1 Explain the meaning of each of the following terms and explain why such species are given conservation priority.

[8 marks]

EDGE species

Endemic species

Keystone species

Flagship species

0 4 . 2 Explain why the degree of population dispersal is considered when setting priorities.

[2 marks]

Do not write outside the box

10

0 5 . 1 Describe **two** case studies of biological control, one successful and one that caused problems.

[4 marks]

0 5 . 2 Explain how pheromone traps can be used in pest control.

[2 marks]

0 5 . 3 Explain why the use of antibiotics as growth promoters in livestock is a concern for human health.

[3 marks]

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0 5 . 4 Give **one** way farmers can support crop pollinators.

[1 mark]

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10

0 6 . 1 Which of the following features of a plantation would most increase its wildlife value? Tick one box.

[1 mark]

A Close planting of a single non-indigenous species ☐

B Mixed indigenous species with a mixed age structure ☐

C Clear-felling on a 40-year rotation ☐

D Removal of dead wood ☐

0 6 . 2 Describe **four** features of sustainably managed forests and explain how each benefits biodiversity or productivity.

[4 marks]

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0 7 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

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2 .....

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0 7 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

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0 7 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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5

0 8 . 1 Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

A Its mass ☐

B Its magnetic field ☐

C Its distance from the Sun ☐

D Its speed of rotation ☐

0 8 . 2 Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

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0 8 . 3 Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

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0 9 . 1 State what is meant by **hybrid vigour**.

[1 mark]

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0 9 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

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0 9 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

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2 .....

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1 0 . 1 Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

1 0 . 2 Explain why growing coral reefs help to resist climate change.

[2 marks]

1 0 . 3 Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

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Write an essay on one of the following topics.

1 1 . 1 Agriculture involves the control of abiotic and biotic factors to maximise food production. Discuss how this control can be achieved and evaluate its environmental impacts.

[25 marks]

OR

1 1 . 2 Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS